

ANSWER KEY

Speech Processing Quiz testing · CSC325

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Q1. In assembly language programming for speech processing applications involving the conversi...

' **D. IN AX**

Explanation: The 'IN' instruction is used to read data from an I/O port into a register. In this context, reading the analog audio signal converted by ADC requires using AX as it can hold two bytes of input.

Q2. In assembly language programming for speech processing systems, which instruction is typic...

' **D. MOV AL, 5AH**

Explanation: The MOV instruction is used to transfer data between registers and memory. In the context of loading a byte from memory into an accumulator register like AX or AL in x86 assembly language, 'MOV' followed by specifying which part (AX/AL) you want to move data to and providing the source address enclosed in brackets is commonly used.

Q3. In assembly language programming for speech processing tasks such as filtering noise from ...

' **D. ADDC AX, BX**

Explanation: The 'ADD' instruction in assembly language is used to perform arithmetic addition. When dealing with audio samples for speech processing tasks like noise filtering, the ADD (Add) instruction can be applied directly or indirectly through registers such as AX and BX.

Q4. In assembly language programming for speech processing systems, which instruction is typic...

' **C. MOVEA AX, 0x5678h**

Explanation: The MOVE instruction is used to transfer data between registers and memory. In this case, 'MOVE' with the destination being AX (a general-purpose register) can load a byte of data from an immediate value or address into AL.

Q5. In assembly language programming for speech processing systems, which instruction is typic...

' **B. LOAD LOW(A), [1234H]**

Explanation: This option correctly represents the instruction to load a byte from memory into register AL (the lower part of AX) which is commonly used for operations on speech data.

Q6. In assembly language programming for speech processing systems, which instruction is used ...

' **A. CMP**

Explanation: The CMP instruction is used to compare two values and set the processor flags without changing their contents. It's commonly utilized in conditional branching within assembly language programs for speech processing systems.

Q7. In assembly language programming for speech processing systems, which instruction is typic...

' **C. MOVBL AL, BL**

Explanation: The correct answer is MOVBL (Move Byte from BL to AL), which loads a byte of data into the Accumulator register's lower part. This instruction directly relates to handling individual bytes in speech processing tasks.

Q8. In assembly language programming for speech processing systems, which instruction is typic...

' **C. MOV AL, data**

Explanation: The 'MOV' instruction is used to transfer immediate values or from one register to another. In this case, MOV AH, #data would load a byte into the AL register if data were of an appropriate size.

Q9. In assembly language programming for speech processing systems, which instruction is typic...

' **B. LOAD A, M[0x1234h]**

Explanation: This option correctly represents the assembly language instruction to load a byte of data from memory into the accumulator register (A), which is essential for initializing or processing speech-related buffers.

Q10. In assembly language programming for speech processing systems, which instruction is typic...

' **B. IMUL DWORD PTR [ebp-4],AX**

Explanation: The IMUL instruction is used to multiply two operands and store the result. In this case, it multiplies a doubleword at address ebp-4 with the AX register.

